

Title: Climate outweighs human effects on vegetation assemblage dynamics in the Holocene

Authors

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Abstract

Although there is consensus that human impacts have shaped ecosystems and biodiversity for millennia, the relative roles of climate and human activity affecting the structure of Holocene ecosystems remain poorly understood. At present, there has been no systematic attempt to determine the relative strength of these two drivers at regional to continental scales. Here, we integrate palaeoecological, archaeological, and palaeoclimate data to quantify the relative importance of climate and human activity in shaping Holocene vegetation assemblage dynamics (8.5 – 2 ka BP). By analyzing 1262 fossil pollen sequences, we examine the responses of nine pollen assemblage properties (PAPs), serving as indirect vegetation assemblage changes and, for the first time, assess how relationships amongst these PAPs change over the study period. We then combine 150,000 radiocarbon dates from archaeological sites with palaeoclimate model output to establish that while human activities had localised impacts, climate was the predominant force influencing vegetation assemblage patterns across continental regions during this period. Whilst human impact during the mid-Holocene was present in some regions, the effects of this human impact was not uniform. Our results challenge the view that human actions were a major driver of changes in broad-scale ecological processes throughout the Holocene.

Main Text

Introduction

Human activity has transformed—and continues to transform—Earth’s terrestrial ecosystems^{1–3}. Over the last centuries, exponential growth in land-use intensity, economic growth, and consumption have had impacts on ecosystems in ways that are unprecedented in geological history^{4–6}. Before the last few hundred years, archaeological and palaeoecological evidence shows that humans have inhabited much of Earth’s land for millennia, transforming natural landscapes into anthropogenic biomes through activities such as burning, deforestation, and cultivation⁷, potentially even influencing global carbon cycling and climate change⁸. The extent of human impact has varied over time, driven by migrations, societal development, technological advancements, and increases in population densities, resulting in regionally distinct rates of vegetation transformation¹.

However, whilst the impacts of human activity on Earth’s ecosystems for the past few centuries are well established, the extent and magnitude of this human impact during earlier periods, particularly on vegetation composition and associated ecological processes, remains under exploration⁹. Lyons et al.¹⁰ presented evidence that human activity was so pervasive that underlying ecological characteristics of communities were modified in the middle Holocene (~6 ka BP), arguing that dominant patterns in taxon co-occurrence patterns shifted from facilitative to competitive interactions as a result of human activity (although see the replies by Bertelsmeier & Ollier¹¹ and Telford et al.¹²). Mottl et al.¹³ reported evidence of consistent increases in rates-of-change in pollen assemblages at continental scales since as early as 4.6 ka BP, likely the result of increased human activity and land-use change, although no attribution studies were carried out to confirm this assumption. Recent syntheses of palaeoecological and archaeological evidence have suggested that humans have likely shaped most of nature over the last 12 ka BP (i.e. Holocene), but evidence indicates that this human impact has occurred at different extents and intensities across continental regions^{7,14–16}. Thus, detection and attribution of ecological changes in relation to human activity in the Holocene remains an important area of research if we are to understand the legacy of past drivers of change on shaping present-day ecosystem patterns.

A number of processes can influence the structure and composition of an ecological community

at a given point in time¹⁷. For example, in addition to land-use change and disturbance related to human activity, climate also plays a major role in structuring vegetation communities, particularly on centennial to millennial timescales^{18–21}. Environmental proxies from ice cores and deep-sea records have confirmed that climate fluctuations during the Pleistocene and the Holocene periods correspond with vegetation dynamics at regional and continental scales²². However, the relative roles of climate and human activity in shaping the structure and function of ecosystems, particularly at larger scales (e.g. regional to continental) within the Holocene period, remains poorly understood. At present, there has been no systematic attempt to disentangle the relative strength of these two drivers during such a critical period of Earth system transformation.

Here, we assembled a global dataset based on fossil pollen data, archaeological remains, and past climate variability (see **Methods**) to investigate the effects of both climate and human activity on vegetation assemblage dynamics—as inferred from pollen assemblages—in the Holocene. We standardised 1262 fossil pollen sequences from North America, Central and South America, Europe, Asia, and Oceania (Fig. 1; Extended Fig. 1) and converted them into nine metrics, each of which reflect different properties of past vegetation assemblage change, including composition, diversity and rates-of-change (Table 1). Since these are derived from pollen assemblages, we stress that these are indirect representations of vegetation assemblage change, hereafter defined as pollen assemblage properties²³ (PAPs, see **Methods**).

We then combined our fossil pollen data with archaeological evidence of past human activity based on two sources: (i) summed probability density (SPD) from over 150,000 archaeological radiocarbon dates²⁴, a validated proxy that goes beyond mere presence-absence indicators by reflecting the intensity and frequency of human activities in a given region^{25,26} (See **Methods**), and (ii) a series of “human-impact events”, derived from expert-based detection of patterns of anthropogenic indicator taxa in pollen diagrams (Extended Fig. 2, **Methods**). Note that we chose to focus on period between 8.5 – 2 ka BP, since there is a lack of radiocarbon datasets for the last two millennia in Europe and Asia²⁴ (Table 2). Finally, we used the *CHELSA-TraCE21k Paleoclimate database*²⁷ to extract climate predictors related to summer and winter precipitation, mean annual temperature, and minimum temperature of the coldest month.

We used this data compilation to test two hypotheses. The first hypothesis (H1) proposes that human impact was the primary driver of changes in PAPs at continental and subcontinental scales between 8.5 and 2 ka BP. We assessed the relative importance of human impact compared to cli-

mate, both spatially (by comparing fossil pollen sequences within climate zones and continents) and temporally (by examining trends across time slices). If human impact is an important driver of change in past patterns in PAPs, we expect its importance to match or exceed the importance of climate across both spatial and temporal dimensions. The second hypothesis (H2) proposes that human impact significantly altered the relationships among the different PAPs within these regions during the same time-period. If human impact was pervasive in the Holocene, we expect that they altered the covariance between the PAPs within different climate zones on continental regions.

Results

H1: Holocene human impact secondary to palaeoclimate

We used hierarchical variation partitioning (HVARPART) within a redundancy analysis (RDA)²⁸ to assess how each predictor—human activities and palaeoclimate—contributes to changes observed in the PAPs, in individual sequences (i.e. Fig. 1, Extended Fig. 3). We find that palaeoclimate has substantially greater importance than past humans in shaping the patterns of PAPs at continental and subcontinental scales (Fig. 2). Our spatial analysis shows high variation in the ratios of predictor importance of past humans and palaeoclimate across the middle to late Holocene (Fig. 2). While both have influenced the patterns in the vegetation metrics at individual sequences—with ratios of importance ranging between 0 and 1—the relative importance of climate increases with spatial scale. For example, at the climate-zone level, weighted mean ratios range from 0–0.71 for humans and 0.29–1 for palaeoclimate. At continental scales, these ratios are 0.15–0.32 for humans and 0.68–0.85 for palaeoclimate. Although palaeoclimate dominates, the impact of past humans is more pronounced in certain climate zones (not in the Polar and Cold climate zones) than its mean importance across all continents. Findings from Oceania, based on fewer than five sequences, lack robustness and should be interpreted with caution.

In addition to assessing the overall contribution of human impact and climate to the patterns in PAPs, we conducted a temporal analysis using both the summed probability densities (SPD) or human-impact events as predictors, to investigate how the importance of these two variables changed over time. Here, we find that palaeoclimate consistently emerges as the predominant factor influencing temporal variations in the spatial patterns of the PAPs across continents

(Fig. 3). Trends are generally consistent among the different regions, though some exceptions exist. In Asia, the importance of humans as indicated by SPD is probably underestimated due to limited radiocarbon dates after 3.5 ka BP, whereas the human-impact events delimited by experts indicate increasing human influence. In Europe, the influence of humans is similarly underestimated after 3 ka BP, although the level of influence estimated by the human-impact events remains relatively stable after this time (Fig. 3). The weighted mean ratios comparing the influence of human population (SPD) and human-impact events to palaeoclimate are presented in Extended Table 3 and Extended Table 4.

H2: Assessing the predictor importance for changes in interrelationships between PAPs

Our second hypothesis explores the stability of relationships among PAPs, over time. Specifically, we aim to determine whether changes in these relationships can be attributed to human activity or climate changes, thereby assessing the drivers behind the observed ecological dynamics. We used principal component analysis (PCA) and symmetrical Procrustes sum-of squares ($m2$) to quantify the change in covariance among the PAPs between each pair of time slices within the climate zones²⁹ (**Methods**). This results in pairwise $m2$ distance matrices that are subsequently used as response variables in a second hierarchical variation partitioning (HVAR-PART) analysis (**Methods**). Mean trends of humans and climate within different continental climate zones, derived by hierarchical generalised additive modelling (HGAM)³⁰ are integrated as predictors (Extended Fig. 4, Extended Fig. 5, Extended Fig. 6, Extended Fig. 7, Extended Fig. 8). The results are summarised in Fig. 4, which show how relationships between our different PAPs evolve over time, stratified by climate zones and continents. Starting at 8.5 ka BP, one can trace the evolving relationship between PAPs for a given climate zone by assessing how its trajectory moves through the ordination space in two dimensions. A stable relationship results in a trajectory spanning regions of the ordination space closer to the origin, while trajectories that span a larger area of ordination space indicate larger directional changes in the interrelationships among one or several metrics. The direction of the arrows illustrates how temporal changes in the interrelationships among PAPs correlate with human or climatic influences, whilst the length represents the strength of these influences, with longer arrows signifying greater correlations.

Based on this analysis, we find notable differences in the importance of humans as predictor for these changes in different continents, represented by the ratios of importance bars (Fig. 4),

but observe that palaeoclimate plays a stronger role in most climate zones. Notably, the coldest regions, particularly Europe and North America's Polar, Cold, and Temperate without dry season zones, show considerably less change in the interrelationships of PAPs compared to other climate zones. This may suggest a climatic filtering effect on the PAPs, with underlying ecological processes either remaining unchanged or changing synchronously throughout the investigated period. In contrast, regions with higher environmental variation—in Cold, Temperate, Tropical, and Arid zones—show more dynamic change (i.e. time trajectories spanning larger areas of the ordination space and evolving in different directions), mainly caused by climatic variability, but also influenced in some regions by varying degrees of past human impact. Interestingly, this analysis demonstrates that past human activities correlate with different climate variables across various climate zones, as evidenced in Fig. 4. Predictor variables that point in the same direction along the gradient indicate positive correlations. We observe that the arrows representing climate variables often point in directions that are distinct from those representing the human predictor, and there are no clear patterns between the different climate zones and continents.

Discussion

Our study used palaeoecological, archaeological, and palaeoclimate data to investigate two hypotheses about the Holocene extent of human impact on past vegetation assemblage dynamics (8.5 and 2 ka BP.). Our first hypothesis proposed that human impact was the primary driver of changes in PAPs, at continental and subcontinental scales between 8.5 and 2 ka BP. Our second hypothesis proposed that the interrelationships between these PAPs were affected by human activity between 8.5 and 2 ka BP. However, while our data show that it is likely that past humans have acted as strong disturbance factors at specific sites affecting establishment, abundance, and succession of past taxon populations, their overall importance is considerably lower than that of palaeoclimate changes. We also find that the influence of human activity consistently increases with decreasing spatial scale. Taken together, we refute both hypotheses and suggest that human impact was not a significant factor in driving vegetation assemblage dynamics at continental to global scales in the study period.

This finding contrasts with a number of previous studies that have identified evidence of humans dominating vegetation and land-use change in the Holocene^{15,31–34}. One possible explanation is that, in our studied time window, prehistoric human populations were likely small and local, and

therefore regional pollen rain masks any small local effects on vegetation in the landscape. Thus, although humans were most likely present across all regions, population densities and land-use type and intensity varied spatially¹, and so the timing and extent of human-impact events also varied considerably between individual sites. As a result, these local human impacts were less significant than any abiotic filtering reflected by climate, which remained the most important predictor in our analysis at continental to global scales.

We also note that, whilst other studies have focussed on land-use changes based on reconstructions of either landcover or anthropogenic biomes^{32,35}, our study goes beyond previous reconstructions of land-use by using PAPs. These PAPS reflect a broader set of vegetation assemblage processes (Table 1) and may not necessarily be correlated with land-use change typically inferred from pollen assemblage composition (e.g. changes in composition do not necessarily imply alterations in species richness). Furthermore, whilst a number of studies have assessed variations in individual PAPs across regions^{e.g. 14,23,31,36}, ours is the first to provide a more holistic perspective, using a multivariate framework to analyse all nine metrics of past vegetation assemblage changes and their interrelationships influenced by humans and climate simultaneously. Such multivariate analyses of PAPs represent a new approach in palaeoecology and more research is needed to understand how variations in these different assemblage properties are influenced by different types of human land-use and other factors.

We emphasise that our findings do not negate the effects of human activity in the Holocene. For example, our analysis indicates that whilst high impact-events, such as total forest clearance, do not show a consistently increasing pattern across regions, certain subcontinental regions do show a consistent increase in the extent of human impact during the period investigated, particularly the first impact of anthropogenic activity and the start of agriculture (Extended Fig. 9). Similarly, whilst our results did indicate that human impact on PAPs at continental scales is more pronounced after ~5 ka BP, this influence was neither spatially coherent nor consistent. For example, human importance surpassing palaeoclimate is observed in Asia (human-impact events) and Oceania (human-impact events and SPD) at 2.5 ka BP and 2.0 ka BP, respectively (Fig. 3). In contrast, for many other regions (e.g. North America, Central and South America, and Europe), palaeoclimate remains the dominant driver in PAPs throughout the Holocene, despite increasing human importance after ~2.0, 1.0, or 3.5 ka BP, respectively. Thus, our results indicate that the site-specific effect of human impact should not be generalised to a continental or global scale and emphasise the importance of climate in shaping the underlying characteristics

of the ecological assemblages in the study period.

Such results have several implications for understanding the long-term interactions of humans and of both past and present ecosystems. First, a number of studies have argued that Holocene human impacts on vegetation assemblages were extensive, potentially even affecting the general rules that govern the formation of biotic assemblages through time¹⁰. If this was the case, then we would expect differences in both the drivers of the patterns in PAPs (H1) and their associations (H2). Our results do not reveal such differences. While we cannot reject the evidence of human impact locally, we emphasise that at, larger scales, climate has remained a dominant predictor of the patterns in our PAPs between 8.5 and 2 ka BP, and any human impacts that did occur were not large enough to result in widespread and universal ecosystem change. Such findings also support the idea that, whilst human impact during the mid Holocene was present in some regions, the effects of human impact are not uniform and so context dependency remains essential for understanding present-day ecosystems.

Second, our findings align with recent global assessments¹⁴ and may help to explain the absence of coherent tipping points in terrestrial ecosystems. The high heterogeneity among regions, with the spatially variable influence of climatic factors and the interactions among taxa, contributes to highly variable assemblage dynamics both temporally and spatially. Such heterogeneity complicates the detection and quantification of overarching trends at a continental or global scale, as it tends to obscure any consistent patterns.

Third, we caution against a prevailing narrative that human transformation of the terrestrial biosphere is always destructive⁷. Our results are in line with other recent studies that human impact has been present in the Holocene, but interestingly, our results support the idea that the effects of preindustrial societies on vegetation assemblage dynamics were less influential than other drivers at continental to global scales^{37–39}. Numerous studies have highlighted the positive effects of human land-use through indigenous knowledge relating to sustainable land-management^{9,40–42}. Although it is evident that some societies have caused substantial ecosystem changes and even local extinctions in the past, our results indicate that climate was the primary driver of underlying ecological processes at regional to continental scales during these times.

Fourth, whilst our results refute both hypotheses about the direct impacts of human activity on PAPs in comparison to climate, our analysis cannot refute the possibility that climate indirectly influenced ecological processes via human impact, such as the effects on harvest yields, carry-

ing capacities, and consequently, population densities⁴³. This complexity makes it challenging to separate the impacts of human activities from those of climate. For instance, in analyses that do not account for feedback loops between humans and vegetation, such as the feedbacks among drought, fire and human activity, all change might be attributed to climate, even though human impact, such as fire management or grazing, could be equally influential if they are closely coupled to climate change⁴⁴. Our second analysis show correlations between humans and climate in different climate zones. These results may argue for a third hypothesis that human activity, particularly farming, was co-shaped by climatic change⁴⁵. These findings underscore the need for further research to fully understand these complex interdependencies.

Note that because of data limitations, our study focuses on the period between 8.5 and 2 ka BP. At present, the lack of availability of ¹⁴C dating for the last 2 millennia makes human population density reconstructions unreliable, but there is widespread evidence of human impact from other sources from this period^{46,47}. For example, Mottl et al. found continental scale increases in rates-of-change starting between 4.6 and 2 ka BP in different regions, with largest effects in the more recent times. Thus, whilst the impacts of human activity are limited in comparison with climate in our study period, we cannot evaluate the effects of preindustrial societies in the most recent millennia. In some regions, the significance of these effects may be heightened by evolving social cultures and complexities during specific archaeological periods^{7,47}. These are historically documented periods of human introduction of many new plants to different continents⁴⁸, which has been linked to increases in pollen richness estimates and compositional changes³¹. At present, there remains a crucial gap that needs to be bridged between localized effects identified in prehistoric land-use and the larger scale transformations since 1950, related to the Great Acceleration⁴⁹.

Similarly, the climate model data that we use are generally appropriate for characterising mean changes. However, there may be mismatches between these data and pollen data, which are often collected from sequences near ecotone transitions to investigate past environmental changes. Such mismatches can arise because the climate models represent broader regional conditions that may not align perfectly with the specific localities from which pollen data are gathered. Additionally, inherent biases of pollen data, such as varying taxonomic resolution and the relative abundances of pollen types influenced by pollen production and dispersal properties, can also mask changes in the PAPs. These biases are further complicated by factors like vegetation density and structure, affecting the likelihood of pollen reaching a site⁵⁰. To mitigate these issues, our

data processing efforts have been designed to reduce the effect of varying taxonomical resolution, different sizes of pollen sums, and unevenly spaced temporal data points. This processing ensured that the inherent biases in the pollen data are more systematic within subcontinental regions, yielding a similar bias across sequences within the climate zones.

Conclusion

The integration of 1262 fossil pollen sequences, archaeological radiocarbon dates, and palaeoclimate data into a large database has allowed us to examine the relative influences of human activity and palaeoclimate on pollen assemblage properties, from 8.5-2.0 ka BP at the global scale. At global to sub-continental scales, our results show that regional climate variability was more important than human activity in driving major vegetation assemblage patterns. Our results caution against extrapolating localised, intense human impacts to broader spatial scales, as we do not find evidence that these reflect widespread or global trends. Our study reinforces the essential role of interdisciplinary research in disentangling the confounding effects of human and climatic influences on past ecosystems.

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Competing interests

The authors declare no competing interests.

Data and materials availability

The source data are deposited and available for download via Zenodo (DOI: 10.5281/zenodo.11369243; <https://doi.org/10.5281/zenodo.11369243>). See methods for data description. The data are under a CC BY-NC-ND licence, which requires approval from data owners before use. Please contact Xianyong Cao (xcao@itpcas.ac.cn) for use of the Asian fossil pollen data.

All R code and workflows to reproduce the data processing, filtering, data analyses, and figures are available for download via Zenodo (DOI: 10.5281/zenodo.12773214; <https://doi.org/10.5281/zenodo.12773214>)

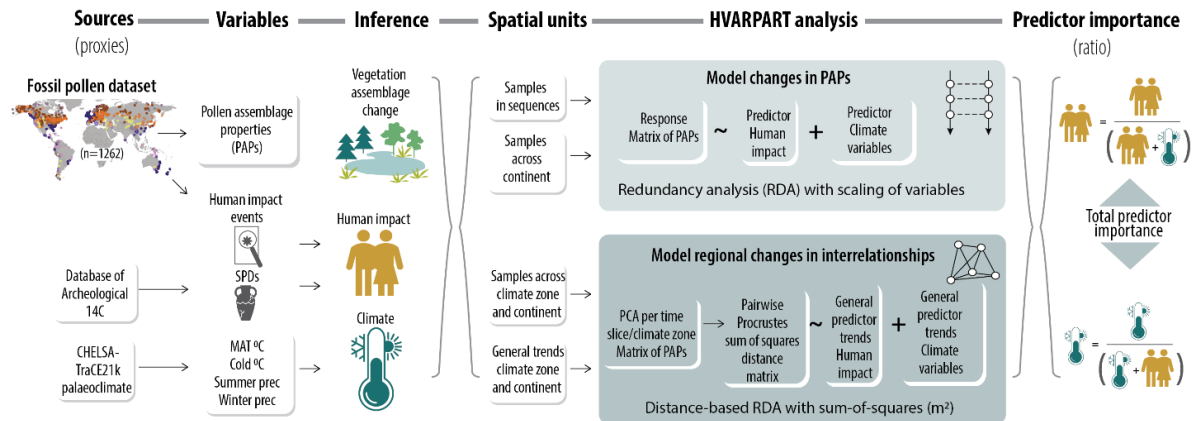


Fig. 1: Conceptual workflow figure for data preparation and quantification of the relative importance of past humans and climate on variability in PAPs at different spatial and temporal scales. From left to right, the main sources of proxy data (Sources (proxies)), what variables are derived from the data (Variables), what they can infer (Inference), data aggregation (Spatial units), the models of the hierarchical variation partitioning analyses and attribution (HVARPART analysis), and the quantification of the ratios of predictor importance between past humans and climate (Predictor importance (ratios)) are shown. See **Methods** for more detailed explanations. Data summaries of fossil pollen sequences, radiocarbon dates, and human-impact events are shown in Extended Fig. 1, Extended Fig. 2. Two site-based examples of quantification of predictor importance between past humans and climate combined are shown in Extended Fig. 3. The general predictor trends of past humans in different climate zones for each continental region are shown in Extended Fig. 4, and the palaeoclimate represented by mean annual temperature (MAT °C), minimum temperature of the coldest month (Cold °C), precipitation of warmest quarter (Summer prec), and precipitation of coldest quarter (Winter prec) for different climate zones are shown in Extended Fig. 5, Extended Fig. 6, Extended Fig. 7, Extended Fig. 8. The general trends for human-impact events in different climate zones are shown in Extended Fig. 9.

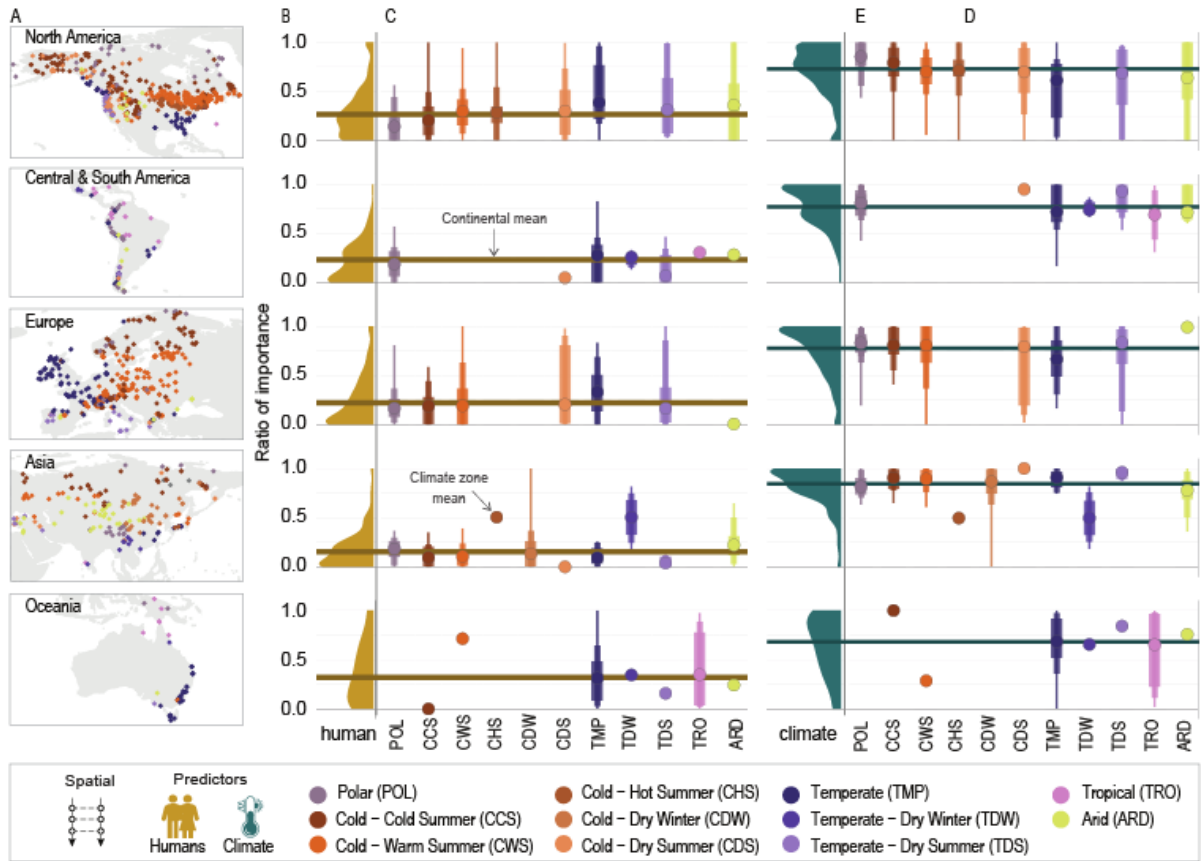


Fig. 2: Ratios of predictor importance of past humans and palaeoclimate on variability in PAPs over the period of 8.5-2 ka BP across climate zones and continents. Each row represents a continent: North America, Central and South America, Europe, Asia, and Oceania. The colours represent Köppen-Geiger climate zones³⁶. Panel A shows the spatial distribution of individual sequences represented by coloured dots on the regional maps (Number of sequences listed in Table 1; Extended Fig. 1). Note that we are only showing the parts of the map with sequences. Panel B shows the regional density curves of the distribution of human importance ratios within a continent (colour gold). Panel C shows the weighted mean ratios of importance (y-axis) of past humans summarised at the level of continents (horizontal gold line) and climate zones (circles). Panel D shows the regional density curves of the distribution of palaeoclimatic importance ratios within a continent (colour teal). Panel E shows the weighted mean ratios of importance (y-axis) of palaeoclimate (colour teal) at the level of continents (horizontal teal line) and climate zones (circles). For each climate zone, the different shading and width of the lines are the percentile ranges (50th, 75th, and 95th) of individual ratios per sequence. Only climate zones with > 2 sequences are used in this visualisation.

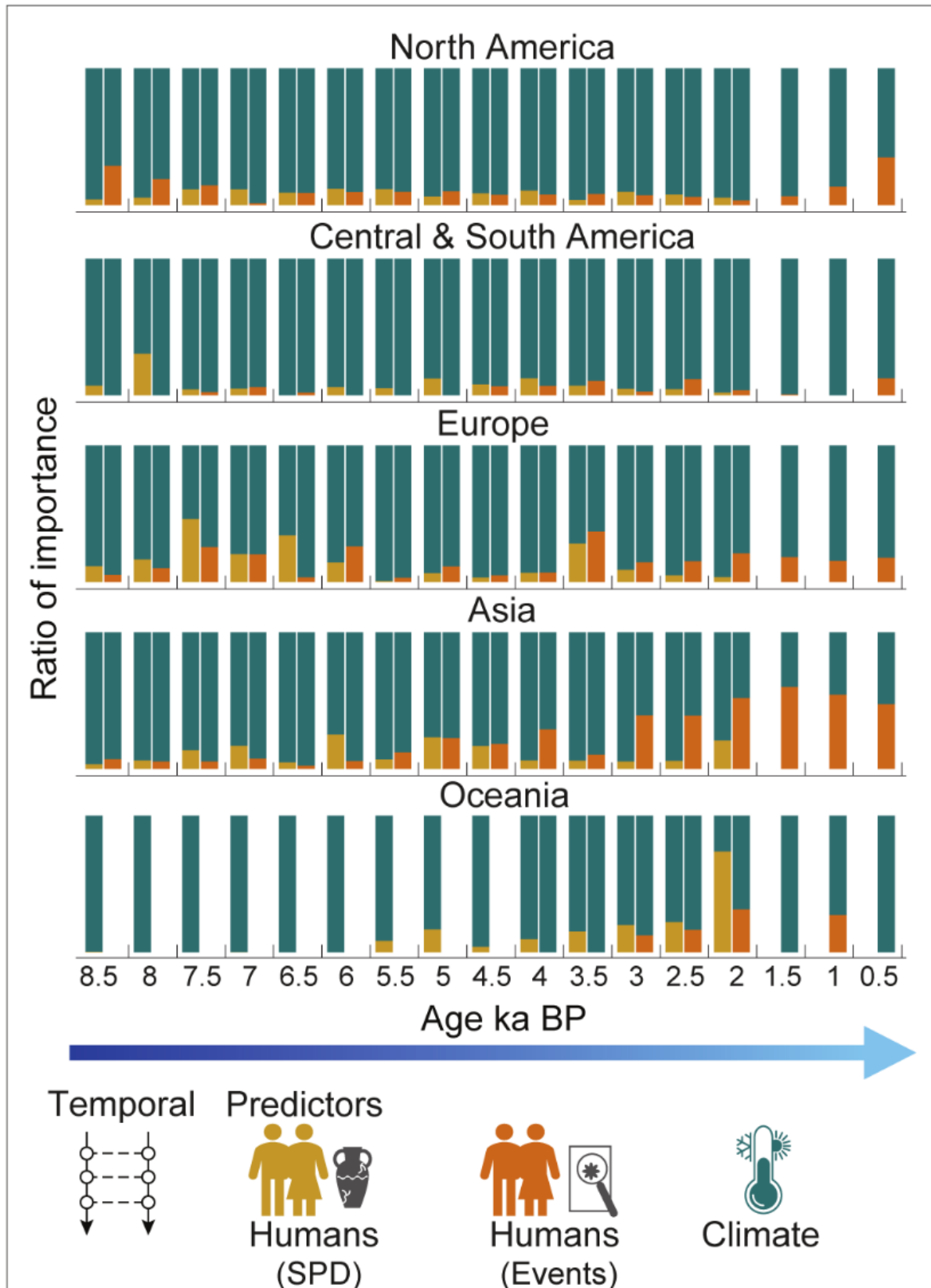


Fig. 3: Ratios of predictor importance of past humans (summed probability density (SPD) and Human-Impact Events) to palaeoclimate on variability in PAPs within time slices for each continent. The bar charts show the ratios of importance between past humans (Humans (SPD)) and palaeoclimate (Climate) as predictors (dark green and gold shaded bars), and the ratios of importance between human-impact events (Humans (Events)) and palaeoclimate (Climate) predictors (dark green and red-brown shaded bars) for the hierarchical variation partitioning (HVARPART). The human-impact events cover additional time slices from 2.0–0.5 ka BP where SPDs are lacking. The ratio of importance (y-axis) varies from 0 to 1.

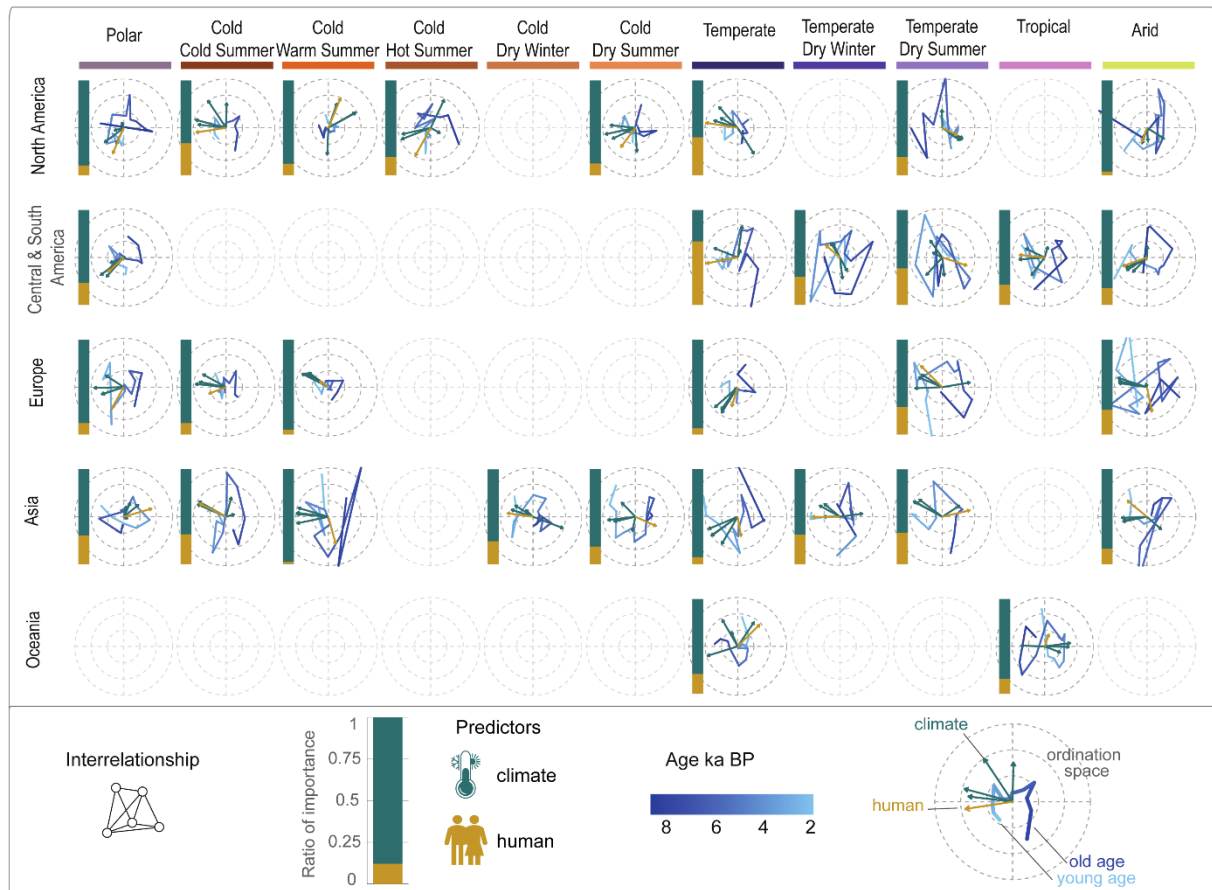


Fig. 4: Ratios of importance of past humans in relation to palaeoclimate in influencing the interrelationships among PAPs in different climate zones across the continents. Each row represents a continental region as follows: North America, Central and South America, Europe, Asia, and Oceania. The columns from left to right represent a total of 11 climate zones within the main Polar, Cold, Temperate, Tropical, and Arid zones. The left bar charts represent the ratios of predictor importance of past human presence (gold) in relation to climate (teal) for changes in the relationships among PAPs for the period 8.5–2 ka BP (y-axis on a scale 0–1). The spread in time trajectories in the ordination space (grey circles) indicates the total amount of change in the interrelationships between PAPs in different time slices, and the ordination space is scaled to similar units to allow comparison across climate zones and continents. The sample scores of the distance-based redundancy analysis (dbRDA) in hierarchical variation partitioning (HVARPART) are plotted as time trajectories (blue lines). These trajectories show the direction of change, where it starts at 8.5 ka BP (dark blue) and progresses towards 2 ka BP (light blue). Arrows, indicating past human presence (gold) and palaeoclimatic variables (teal), show the influence of each predictor over time on the interrelationship among PAPs. The direction of the arrows indicates the trend of influence, while their length quantifies the strength of association between predictors and changes in interrelationships among PAPs, with longer arrows representing stronger influences.

Table 1: Overview of metrics used to estimate PAPs.

Assemblage metrics	Description
N0	Number of pollen types
N1	Number of equally common pollen types
N2	Number of equally abundant pollen types
N2/N1	Ratio of the number of equally abundant pollen types divided by equally common pollen types
N1/N0	Ratio of the number of equally common pollen types divided by estimated number of pollen types
DCCA1	Compositional turnover constrained by time
RoC	Rates of compositional change
Density diversity	Density estimates of break-points in diversity variables N0, N1, N2, N2/N1, N1/N0
Density turnover	Density estimates of break-points in turnover variables estimated by multivariate regression trees (MRT), DCCA, and RoC values

Table 2: Data summary for each continental region. Number of pollen sequences included in the analyses for the assessed period of 8.5–2.0 calibrated thousand years before present (ka BP), the number of sequences with summed probability densities (SPDs) within a 250 km or 500 km radius of the site, and the number of pollen sequences with human-impact events.

Continents	Number of fossil pollen sequences	Number of sequences with SPD 250 km	Number of sequences with SPD 500 km	Number of sequences with human-impact events detected
North America	554	512	43	217
Central & South America	87	65	15	87
Europe	396	383	14	143
Asia	177	47	64	95
Oceania	48	37	1	28

Methods

We created a GitHub repository to host all R scripts and custom-made functions to process data and generate outputs for this project, ensuring full transparency and reproducibility (<https://github.com/HOPE-UIB-BIO/PastHumanImpact>). Additionally, all data, under specific licences, have been uploaded to ZENODO. Specific links are provided under **Data availability**. Our full workflow encompasses all steps from data input to the final outputs, including pollen sourcing and processing, calculations of pollen assemblage properties (PAPs), and statistical analyses.

FOSSILPOL Workflow

Raw pollen datasets were curated using the R-Fossilpol package, following the guidelines in Flantua et al.⁵¹ and detailed on the FOSSILPOL project website^{52,53}. The majority of data sequences were sourced from the *Neotoma Paleoecology Database*⁵⁴, with additional data from private sources used to address regional gaps, particularly Asia. These additional datasets were accessed under CC BY-NC-ND 4.0 licence²³. Extended Table 1 provides an overview of the parameters employed in the FOSSILPOL workflow for compiling and processing of pollen sequences. This workflow was refined during subsequent data-processing stages, yielding a high-quality compilation of pollen sequences suitable for numerical analyses and calculations of PAPs.

Harmonisation tables

A key step in the FOSSILPOL workflow is the taxonomic and nomenclatural standardisation of pollen types to ensure dataset consistency both within and across regions. Variation in identification criteria and nomenclature among analysts from diverse backgrounds and schools may result in substantial discrepancies in pollen taxonomic levels and naming conventions, which would bias numerical comparison⁵¹. To mitigate these biases, we developed harmonisation tables all regions included in this study, encompassing Europe, Levant, Siberia, Southern Asia, Northern America, Central and South America, and the Indo-Pacific region. These harmonisation tables^{23,55}, were incorporated into the FOSSILPOL workflow.

Data filtering

To ensure the highest possible data quality and facilitate numerical comparisons across multiple pollen sequences, several filtering criteria were carefully chosen and applied:

- Removal of potentially duplicated pollen sequences.
- Sorting assemblages (samples) by chronological order (age).
- Exclusion of assemblages (samples) with a total pollen count below a specified threshold (see below).
- Exclusion of pollen sequences falling outside defined age ranges.
- Exclusion of assemblages (samples) based on the age of the last control point.
- Exclusion of assemblages (samples) beyond the specified age ranges of interest.
- Exclusion of pollen sequences if the total number of samples was fewer than 5.

These filtering procedures were systematically applied to the chronologies, raw pollen counts, harmonised pollen counts, and age uncertainties derived from the age-depth models (Bchron)^{56,57}. While the preferred minimum number of pollen grains was initially set at 150, this threshold resulted in a significant loss of pollen sequences, particularly in regions with limited data coverage. To balance data retention and quality the threshold was adjusted to 25, with the condition that less than 50% of the samples would have a low pollen sum. However, it should be noted that in such cases, the estimates of properties of pollen assemblages may exhibit reduced robustness. Regrettably, we excluded Africa as a region from our analysis because at the start of our project (2018) the coverage of pollen sequences available was too sparse. See data summaries in Extended Fig. 1. A complete description of all filtering criteria used in the FOSSILPOL workflow is provided in Extended Table 1.

Estimating Metrics of Pollen Assemblage Properties (PAPs)

We calculated different metrics, derived from community ecology and biodiversity analyses, that reflect different properties of a pollen assemblage^{e.g. 23} for each pollen sequence in our dataset compilation. These are considered proxies for changes in different components of a vegetation

assemblage over time. We note that although there is not a direct relationship between a pollen assemblage and the vegetation assemblage it might represent, the association between these two concepts is fundamental to palaeoecological research⁵⁸. We used a suite of metrics that represents palynological richness, diversity, evenness, compositional change and turnover, and rate-of-change (RoC). These metrics reflect components of an ecological assemblage, caused by underlying ecological processes and interactions between past species populations and environment, which may result in taxon gains or losses (richness), changes in abundances (evenness), change in both richness and evenness (diversity), and functioning and structure of assemblages (composition), and how a pollen assemblage changes over time and space (turnover and rate-of-change).

We calculated most metrics using the newly developed *R-Ecopol* package⁵⁹. The base functions used in this package are derived from other packages. Examples are *mypart*⁶⁰ to derive pollen zonations with multivariate regression trees, and *vegan*⁶¹ for other multivariate techniques and dissimilarity indices.

Pollen richness, diversity, and evenness

Palynological richness, diversity, and evenness were estimated using Hill's effective species numbers N_0 , N_1 , and N_2 , and the associated evenness ratios of N_2/N_1 and N_1/N_0 ^{58,62}. The effective species numbers mainly differ in how rare taxa are weighted in comparison to abundant taxa depending on the parameter q in the formula⁵⁸:

$$D^q = \left(\sum_{i=1}^T p_i^q \right)^{1/(1-q)}$$

where D is the diversity estimate, and p is the proportion of taxon T_i . When q is 0, rare and abundant taxa have equal weight, and the number is simply the number of taxa in the sample. The equation is not defined for $q = 1$, but when it approaches 1, it equals the exponential of the well-known Shannon index and provides the number of equally common taxa. When $q = 2$, it is the same as the inverse Simpson diversity index and is the number of abundant taxa. Rare taxa receive a low weight. The advantage of using effective species numbers is that they provide easily interpretable units and contain the doubling effect⁶². To standardise pollen sample sizes, we used the rarefaction approach developed by Chao et al.⁶². These estimates were rarefied

to the number of $n = 150$ grains, or in a few cases to a lower sum (minimum $n = 25$). Some pollen sequences from Asia were only available as pollen percentages; these are rarefied to the minimum sum of the percentages. The ratios (e.g. $N2/N1$) between any diversity numbers will be 1 if all taxa in a sample are equally abundant, hence these ratios reflect changes in evenness between rare, equally common, and abundant taxa.

Functions from *iNext*^{60,62,63} were modified to extract interpolated Hill numbers based on a minimum sample size and incorporated into *R-Ecopol* package⁵⁹.

Despite our efforts to standardise richness and diversity estimates based on a standard sample size, there remain unaccountable biases, which include variations in total pollen production and pollen representation^{50,64}.

Compositional turnover

We estimated the compositional turnover using detrended canonical correspondence analysis (DCCA) with age as the explanatory variable⁶⁵. Changes in weighted average (WA) scores (*CaseR* scores) are measures of compositional turnover in standard deviation (SD) units⁶⁶. The WA scores were regressed with time using a second-order polynomial ($\text{age} + \text{age}^2$) to allow more flexibility in the turnover pattern within a pollen sequence. The total range of *CaseR* scores along DCCA axis 1 is a measure of total compositional change for a sequence, whereas the pattern of individual *CaseR* scores along DCCA axis 1 is a measure of turnover between individual samples within the sequence⁶². The response data are pollen percentages without any transformation to maintain the chi-square distances between samples that is implicit in DCCA, whereas the ages are the median ages derived from the age-depth model for each sequence. We specifically developed R functions in the *R-Ecopol* package to run DCCA using *Canoco* 4.5⁶⁵.

Rate-of-change

To quantify the rate-of-change (RoC) of composition, we used the *R-Ratepol* package⁶⁷. RoC was calculated using moving windows of 500-year time bins, with five shifts for each bin where samples were randomly selected. This approach, as demonstrated by Mottl et al.⁶⁷, enhances the accurate identification of RoC peak points compared to more traditional methods. The RoC values are reported as dissimilarity per 500 years, employing the chord dissimilarity coefficient⁶⁸.

Standardisation of sample size was performed within each working unit, ensuring consistency at 150 grains or the lowest number detected in each dataset.

Change-point detection and density estimations

For the calculation of change points in all the PAPs, we used conventional regression trees (RT) for single variables with Euclidean distances. An algorithm was developed to identify transitions between resulting groups or zones per variable by coding these transitions as binary variables (0/1). A change point was designated as 1, with the mean ages between two consecutive samples representing the timing of this significant change, determined individually for each pollen-assemblage property.

Compositional change was calculated using multivariate regression trees (MRT) with age as the sole constraining variable. MRT is, in general, a robust tool to explore and predict changes in multivariate datasets using external predictor variables^{68,69}. This technique has been adopted in palaeoecology to detect zones in pollen diagrams or shifts between periods of homogeneous assemblages in time⁶⁸. We used the pollen taxa in percentages without any data transformations as the response variable and the median ages derived from the age-depth model as the constraining variable. The recursive partitioning is based on chi-square distances⁷⁰ between pollen samples constrained by time. The number of cross-validations was set to 1000, and the optimal sized tree was chosen based on the 1SD rule⁶⁸.

Significant change points for richness, diversity, and evenness variables were integrated into one variable (DDC), while significant change points for compositional turnover, compositional change, and rate-of-change were combined into a second variable (DCC). The density of these two variables was computed using a Gaussian kernel and rescaled to specific age ranges for each individual pollen sequence, including minimum and maximum ages. To address boundary concerns in density estimation, the data were reflected to 0.

Equal temporal spacing of samples

We ensured equal temporal spacing of samples across all response variables, as this is crucial for temporal analyses in our stated hypotheses 1 and 2. This involves standardising samples to intervals of 500 years across time and space. Three methods of interpolation were tested: generalised

additive model (GAM), hierarchical generalised additive model (HGAM), and simple linear interpolation. Linear interpolation was ultimately chosen as the preferred method, as it maintained the correlation structure between multivariate response variables closest to the original estimates, compared to GAM or HGAM. The latter models occasionally exhibited unexpected patterns in single pollen diversity estimates, which affected these correlations. Given the impracticality of individually assessing each model for every variable across all sequences (numbering over 1000), we opted for linear interpolation as the simplest approach.

After all data-processing steps, a total of 1262 sequences were available for analyses. For an overview of data coverage, see Extended Fig. 1.

Spatial units for analysis

The analyses focused on three levels of spatial units: 1) individual sequences, 2) sub-regions within continents, and 3) continental regions. While we acknowledge that any grouping will be imperfect and some specific regions might be less well represented, we used present-day Köppen-Geiger climate zones⁷¹ as a classification at the subcontinental scale. The advantage of using the Köppen-Geiger climate zones lies in its reliance solely on climate data, without making assumptions about vegetation or human influence. Additionally, these climate zones are characterized by relatively stable climates within each zone, enhancing the detectability of potential human impacts.

We used five major climate zones, namely Polar, Cold, Temperate, Arid, and Tropical for visualisation, but since the Cold and Temperate zones are so widespread across the Northern Hemisphere, we sub-divided these into Cold without Dry Season & Very Cold Summer, Cold without Dry Season & Cold Summer, Cold Dry Winter, Cold Dry Summer, Cold without Dry Season & Warm Summer, Cold without Dry Season & Hot Summer, Temperate without Dry Season, Temperate Dry Winter, and Temperate Dry Summer. This resulted in 12 climate zones used in the analyses.

Detection of past human activities

To assess the historical influence of past human activities on past vegetation (i.e., pollen-assemblages), we created two proxies of past human presence for each sequence: (1) summed probability densities (SPDs)²⁴ of archaeological radiocarbon (14C) dates to quantify past human population and their regional dynamics, and (2) human events derived from fossil pollen diagrams based on expert knowledge. We used both proxies to assess the importance of past human presence and impact across regions and continents (Extended Fig. 2). Such integration also helped to mitigate the potential issue of circularity wherein human events detected from the same pollen sequences are used to estimate ecological attributes of the pollen assemblages.

Past human presence from archaeological radiocarbon dates

We used a global dataset of archaeological radiocarbon (14C) dates from Bird et al.²⁴ to construct SPDs, which serve as validated proxies for regional human population changes²⁶. To associate radiocarbon dates with pollen sequences, a specific radius around each pollen sequence was defined. The default radius was set at 250 km to capture a reasonable density of 14C dates, with the radius increased to 500 km for sites with insufficient coverage. This approach assumes that human populations were present and potentially influencing the region within these distances

Radiocarbon dates were carefully filtered for validity, ensuring accurate geographical location (longitude and latitude), and ‘LocAccuracy’ > 0. These refined 14C dates were calibrated using the “*calibrate()*” function from the *rcarbon* package⁷². Appropriate calibration curves (“IntCal20”, “ShCal20”, “mixed”^{73,74}) were applied, with the mixed curve generated using the “*rcarbon::mixCurves()*” function, setting $p = 0.5$. Calibration curve assignment was based on geographical location, following Hua et al.⁷⁵. The estimation of SPD was performed using the “*spd()*” function from the *rcarbon* package ranging from present to 12 ka BP. Sequences with fewer than 50 radiocarbon dates were excluded to maintain robust SPD estimates. We used a moving window of 100 years to smooth the non-normal peaks.

The number of 14C dates decreases strongly toward younger material, particularly over the last 2 ka BP in Eurasia²⁴. Therefore, we focused on the time-period between 8.5 and 2 ka BP, ensuring sufficient radiocarbon data and reliable SPD reconstruction. SPD values were extracted at every 500-year interval, providing a time-resolved predictor of past humans representing the

magnitude of regional changes in human populations in different regions.

Human events derived from fossil pollen sequences

We employed expert-based approach to detect periods of human activity in each pollen sequence. This binary approach identified “human events” based on anthropogenic indicator taxa, allowing us to integrate supplementary predictors of human presence and activity.

Detection of human events in North America, Europe, Asia, and Oceania

For these regions, human events were identified through examination of each pollen diagram sequence by one or more regional specialists, who recorded the age of specific events (i.e., changes in anthropogenic indicator taxa). Event types were regional-specific and may have different interpretations even if the taxon involved is distributed across continents (see Extended Table 2). A binary matrix (0/1) was created for each event type detected in each pollen diagram sequence, incorporating the average ages between levels (samples), considering that human impact may have occurred in the intervals between samples in the fossil pollen sequences. Binary values (0/1) were assigned to different event types based on their presence (i.e., detection and the event’s age). This assignment adhered to logical principles, and binary values were adjusted to compile event data for each pollen sequence. This results in a binary matrix of ages and event types for each sequence.

Note that the absence of human events in a pollen diagram sequence does not necessarily imply the absence of humans but rather indicates insufficient evidence to confirm human activity as detected in the pollen sequence.

Detection of human events in Central and South America

Due to the complex and unique use of plants in Central and South America, we performed a thorough literature review targeting pollen types associated with human presence and/or activity⁴⁸:

1. Anthropogenic pollen indicators were identified from the literature as single human indicators or combined variations in a group of indicators (i.e. “human indices”).

2. For each pollen sequence, the combinations of taxa recorded alongside specific indicator groups was considered. We evaluated whether the signal of a pollen indicator for human presence was weak or strong, applying a defined threshold.
3. Two categories were established: “human-event indicators”, representing a single taxon associated with human activity categorised as having “weak” or “strong” impact; and “human-event indices”, comprising combinations of taxa classified similarly.
4. For each pollen sequence, the human-event indicators and human-event indices were assessed for their presence at each level, with classification as “no impact”, “weak impact” or “strong impact”. Specifically, the presence of a single pollen grain of a human indicator signified “weak” impact, while the presence of more than one pollen grain of a strong impact indicator denoted “strong” impact. Additionally, the pollen type *Pinus* was considered a human indicator outside its native distribution, and for human-event indices (step 3), any quantity of pollen grains had to be present for the specific combination of taxa to be classified.

Palaeoclimate

Palaeoclimate from the CHELSA-TraCE21k downscaling algorithm was downloaded from the CHELSA database which is based on the Community Climate System Model version 3²⁷. The selected bioclimatic variables are *bio1*: annual mean temperature (°C), *bio6*: minimum temperature of the coldest month (°C), *bio18*: precipitation of warmest quarter (kg m⁻² quarter-1, = mm), and *bio19*: precipitation of coldest quarter (kg m⁻² quarter-1, = mm). The variables were selected *a priori* to represent subcontinental climate variability with respect to differences in temperature, moisture, or seasonality between the continental regions. The climate values were extracted for the co-ordinates for each *dataset_id* (pollen sequence) to retrieve the full time series for every 100 years for each climate variable. The point-based extraction of co-ordinates may provide a mismatch in elevation in areas with rough terrain. However, we assume the regional extent of the pollen signal is fairly comparable to the relevant spatial area of the climate signal per 1km spatial grid.

Numerical analysis of hypothesis 1

To calculate the relative importance of human events and climate for variability in PAPs, we used hierarchical variation partitioning (HVARPART) developed by Lai et al.²⁸. This approach calculates the importance of each predictor by partitioning the unique and shared variation for all possible combinations of predictor orders in a full model. The predictor importance is then the sum of the average unique and shared variation²⁸. To obtain an estimate of predictor importance, we used redundancy analysis (RDA) with standardised response variables as pollen assemblage properties are of mixed units. Past humans (e.g. SPD values) represent one predictor, whereas climate is represented as a group of selected variables of summer precipitation, winter precipitation, annual temperature, and winter temperature. We used the adjusted r^2 total predictor importance and the average unique partitions as they correct for the number of samples and predictors in the model. To compare the predictor importance of humans and climate, we estimated the individual importance ratios for each sequence as follows:

$$\text{Predictor importance ratio} = \text{predictor importance} / (\text{predictor importance human} + \text{predictor importance climate})$$

The analysis was run in two ways:

- e. to analyse the *spatial changes*, HVARPART was run for individual sequences. To summarise the results on broader scales within the continents, we used the Köppen-Geiger climate zones to categorise subregions and calculated the weighted mean ratios of the total predictor importance from each sequence using the sum of the predictor importance of humans and palaeoclimate as weights.
- f. to analyse the *temporal patterns* for each continent, HVARPART was run for samples within each 500-year time slice. Time slices with fewer than five samples were excluded, and some analyses failed when SPDs indicated zero human presence. Missing results in these cases were attributed to insufficient predictors. The predictor groups in each time slice consist of the SPD or group of human-detection events (Extended Table 2) together with the matrix of palaeoclimatic variables.

HVARPART analyses were run using the *rdacca.hp* R package²⁸. Two site-based examples to analyse spatial changes are provided in Extended Fig. 3.

Numerical analysis of hypothesis 2

To test if the interrelationships between PAPs have changed over time (H2), we performed a principal component analysis (PCA) with standardisation of the metrics within each 500-year time slice. Here samples were grouped at the subregional levels of climate zones within each continent. We performed a Procrustes sum-of-squares analysis to compare the changes in the interrelationships between PAPs between time slices²⁹. We used symmetric scaling (*m2*) creating distance matrices representing these changes, where 0 indicates the relationship remained constant, and 1 indicates that the relationship between one or several PAPs has changed considerably. Each climate zone is represented by the *m2*-distances between time slices for the period 8.5–2 ka BP. This was used as a response matrix in HVARPART.

Bayesian modelling was employed to establish overarching trends in past humans (SPD) and palaeoclimate³⁰. It is important to note that the predictions generated by these models encapsulate the prevailing trends in past human presence and palaeoclimate as relevant to the distribution of our compiled sequences and these patterns may not directly represent the broader historical human presence and palaeoclimate conditions in various climate zones; see Extended Fig. 4, Extended Fig. 5, Extended Fig. 6, Extended Fig. 7, Extended Fig. 8. Subsequently, these variables were employed as predictors in the second HVARPART, aimed at quantifying the importance of past human or palaeoclimate to the variability in the interrelationships between PAPs within climate zones. The general trends were also modelled for the human-impact events in different climate zones across the continental regions (Extended Fig. 9).

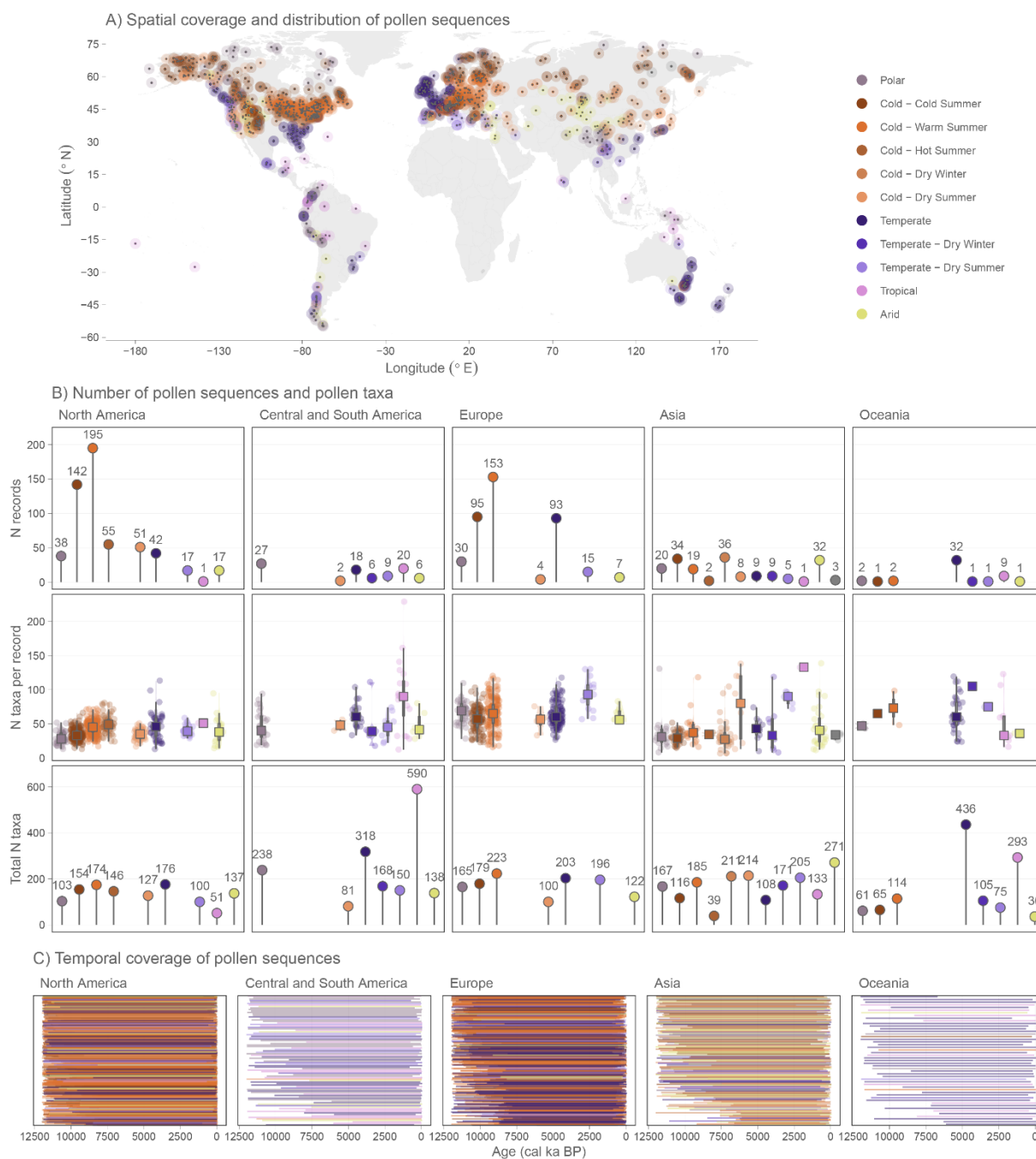
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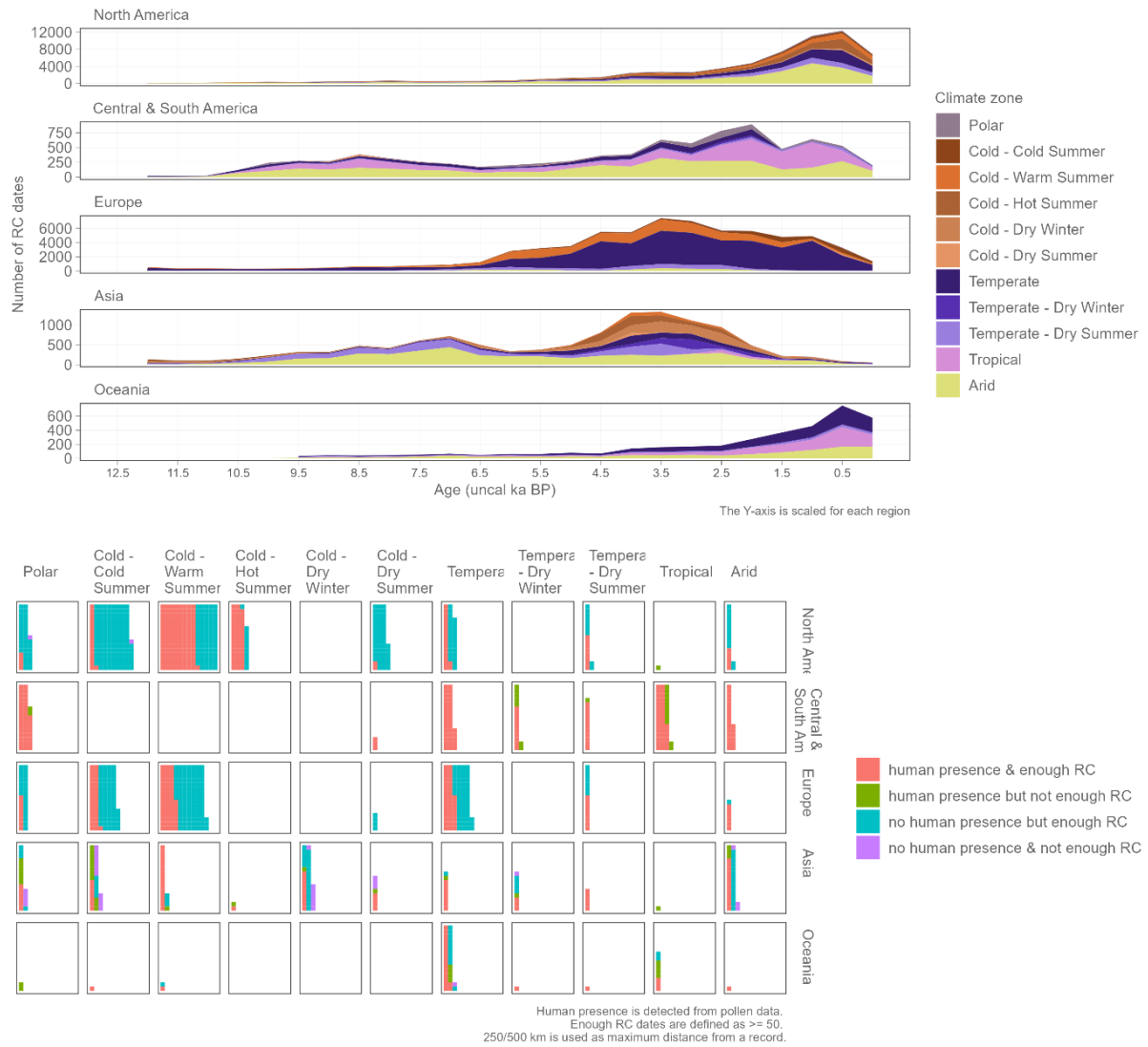
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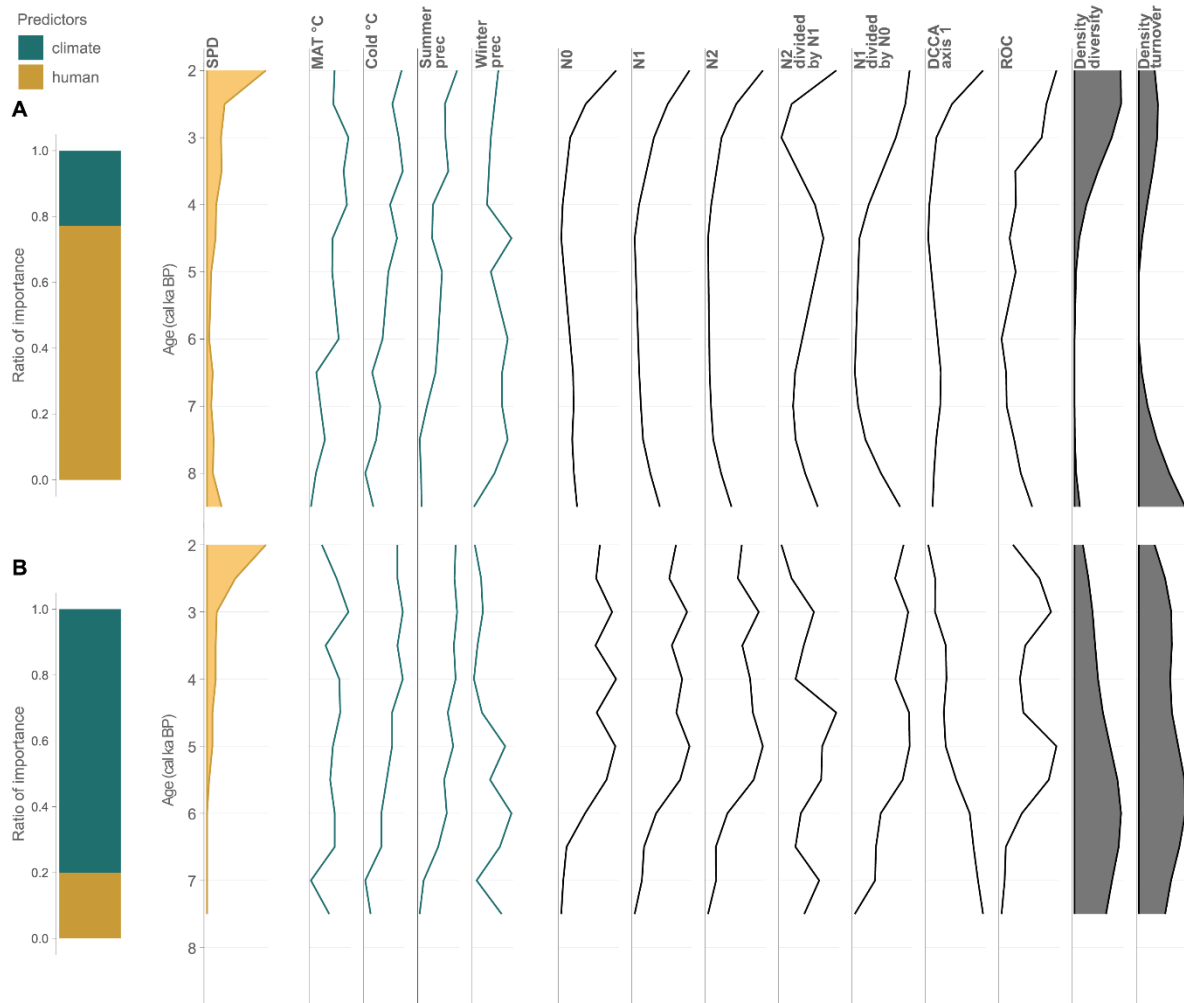
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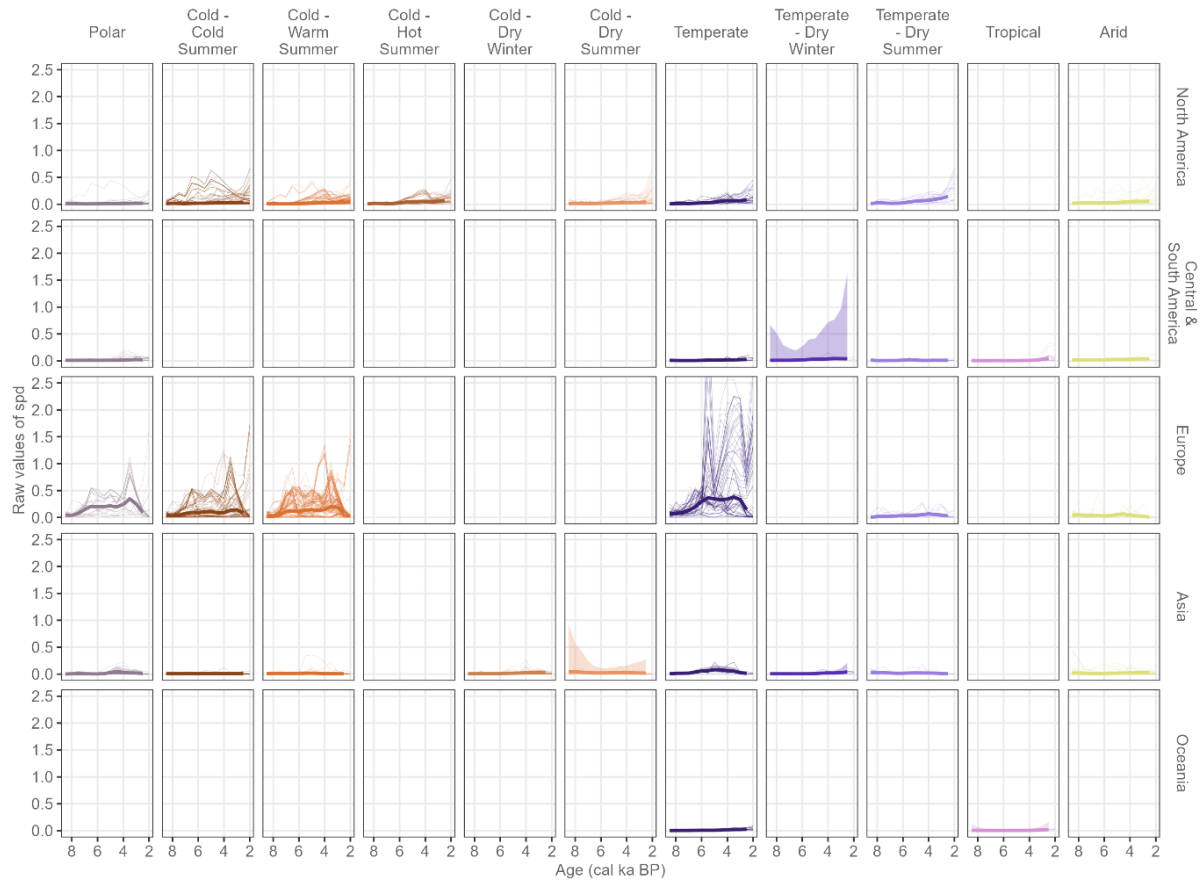
Extended Fig. 1: Overview of data coverage. (A) Spatial coverage and distribution of pollen sequences, (B) number of pollen sequences within climate zones (panels row 1), the number of taxa per sequence in each climate zone (panels row 2), and total number of taxa in each climate zone (panels row 3), and (C) temporal coverage of the pollen sequences.



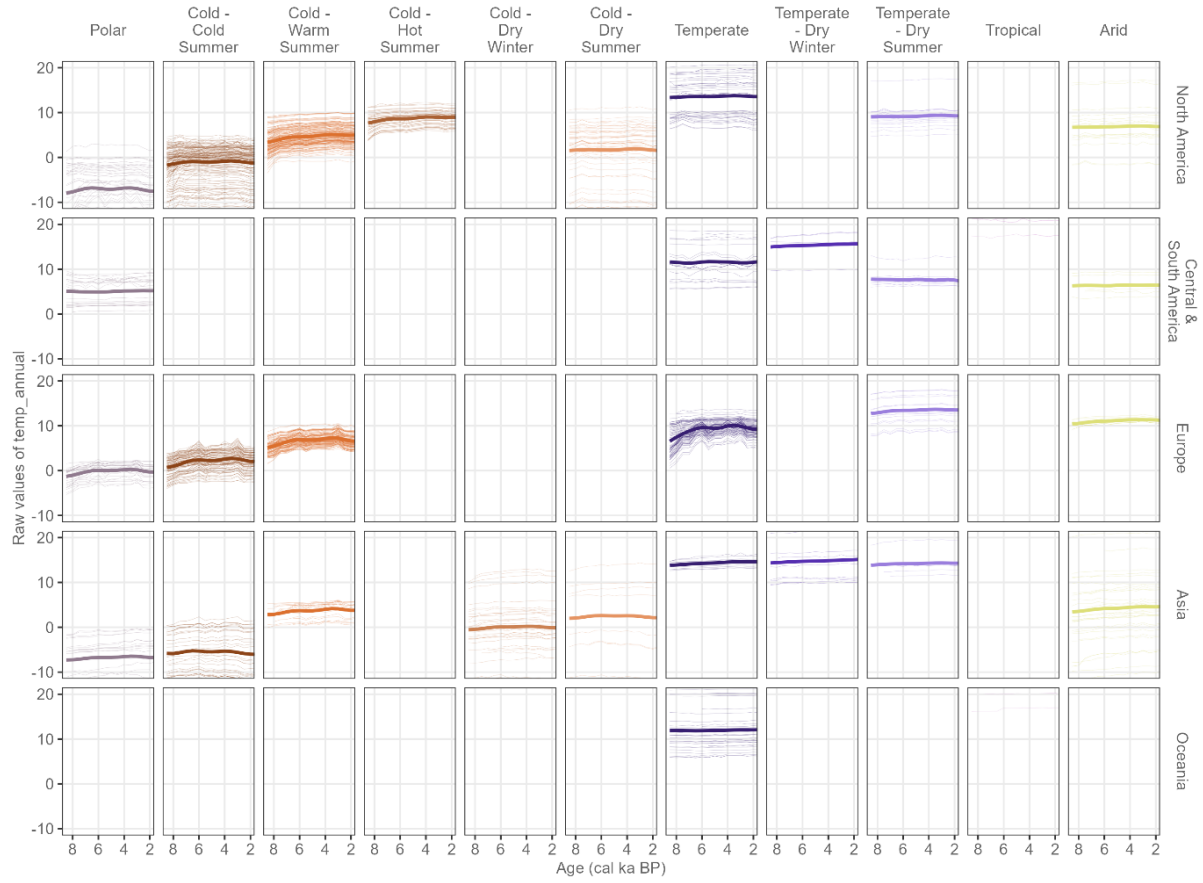
Extended Fig. 2: The top panel shows the number of uncalibrated radiocarbon (RC) dates for each climate zone in the continental regions. The y-axis, scaled for each region, is the number of RC dates. The bottom panel shows a comparison of data coverage of human presence detected from pollen diagrams and radiocarbon dates ordered by climate zones (right to left) and geographical regions (top to bottom). Am = America



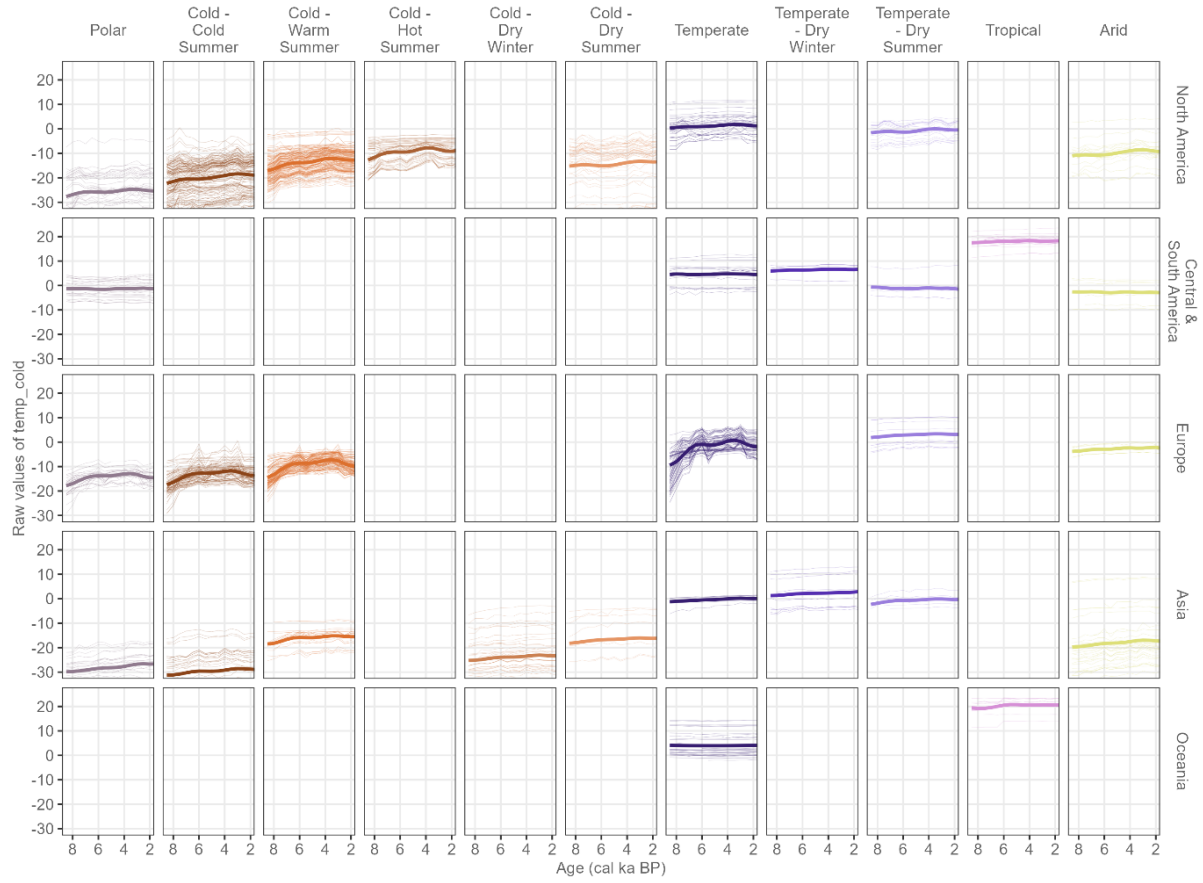
Extended Fig. 3: Two sequences and their temporal trends of PAPs and predictors (columns). The bar plots show results of the hierarchical variation partitioning. The two sequences were selected as examples of contrasting results: A) PAPs are mostly affected by trends of human population (SPD); B) PAPs are mostly explained by climate (MAT °C, Cold °C, Summer prec, Winter prec). The calculated metrics from each sequence of our full data compilation are subsequently used as input data in the analyses summarised in Fig. 2. SPD = summed probability densities, MAT °C = mean annual temperature, Cold °C = minimum temperature of the coldest month, Summer prec = mean precipitation of the warmest quarter, Winter prec = mean precipitation of the coldest quarter, N0–N2 = Hill’s diversity numbers, N2/N1 and N1/N0 = Hill’s evenness numbers, DCCA axis 1 = first axis of detrended canonical correspondence analysis, ROC = rate of compositional change, Density diversity = density of change points for diversity variables, Density turnover = density of change points of turnover variable. See **Methods** for more details about estimation of each variable.



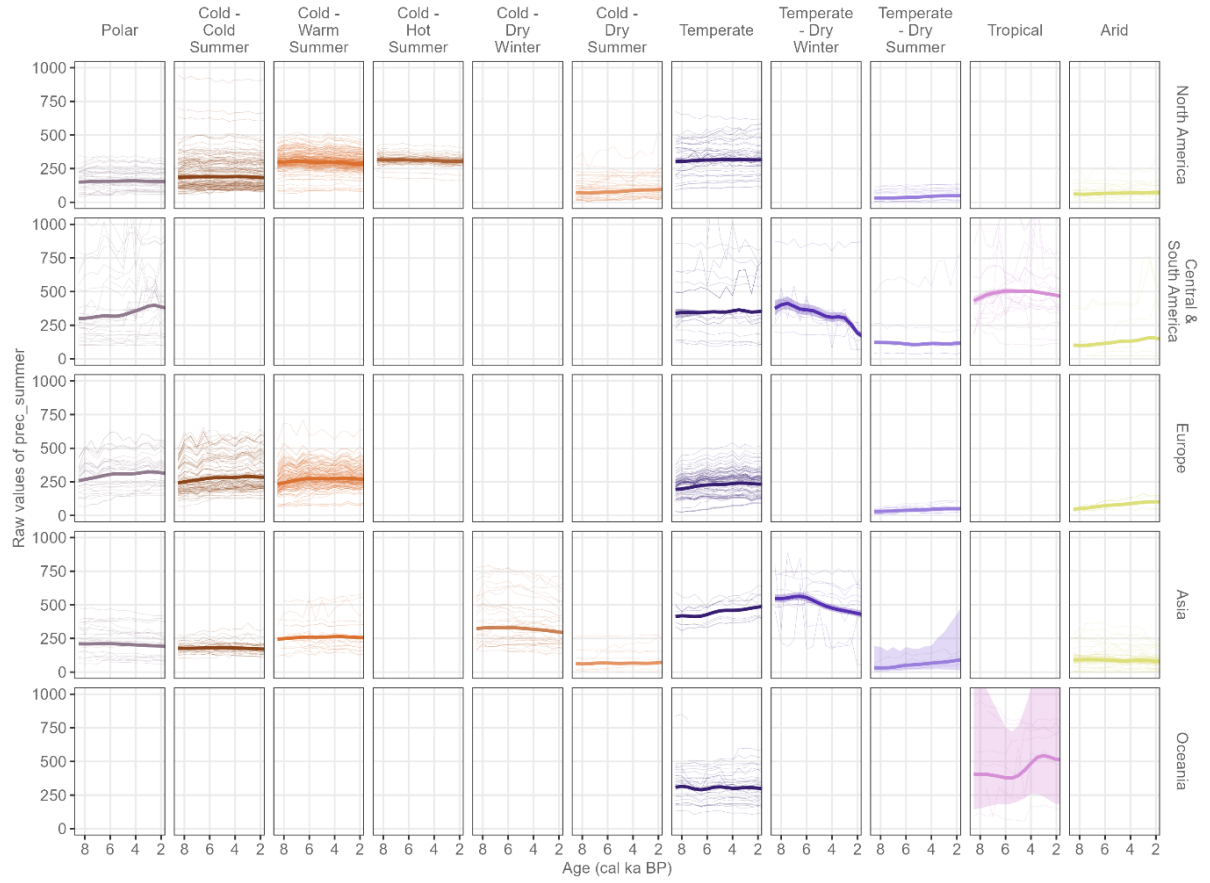
Extended Fig. 4: General patterns of the summed probability density (SPD) curves based on all sequences within each climate zone for each continent. The thin lines represent the raw data, while thick lines represent the predictions from the hierarchical generalised additive models (HGAMs). Note that values have been normalised and centralised for this visualisation.



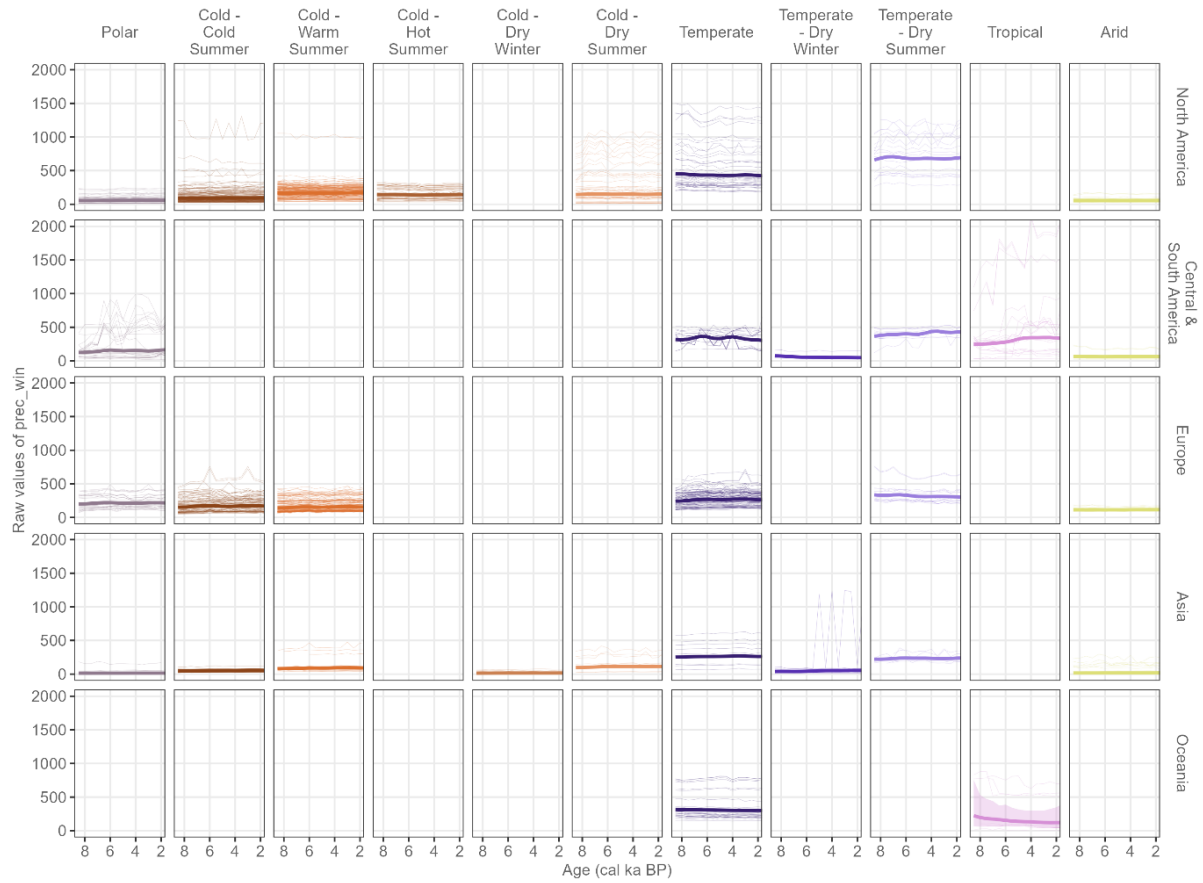
Extended Fig. 5: General patterns of mean annual temperatures °C (*bio1*) in each climate zone and continent, derived from CHELSA climate data²⁷. Thin lines represent different trends in the raw data at the sequence level. Thick lines show the general prediction from the hierarchical generalised additive models (HGAMs).



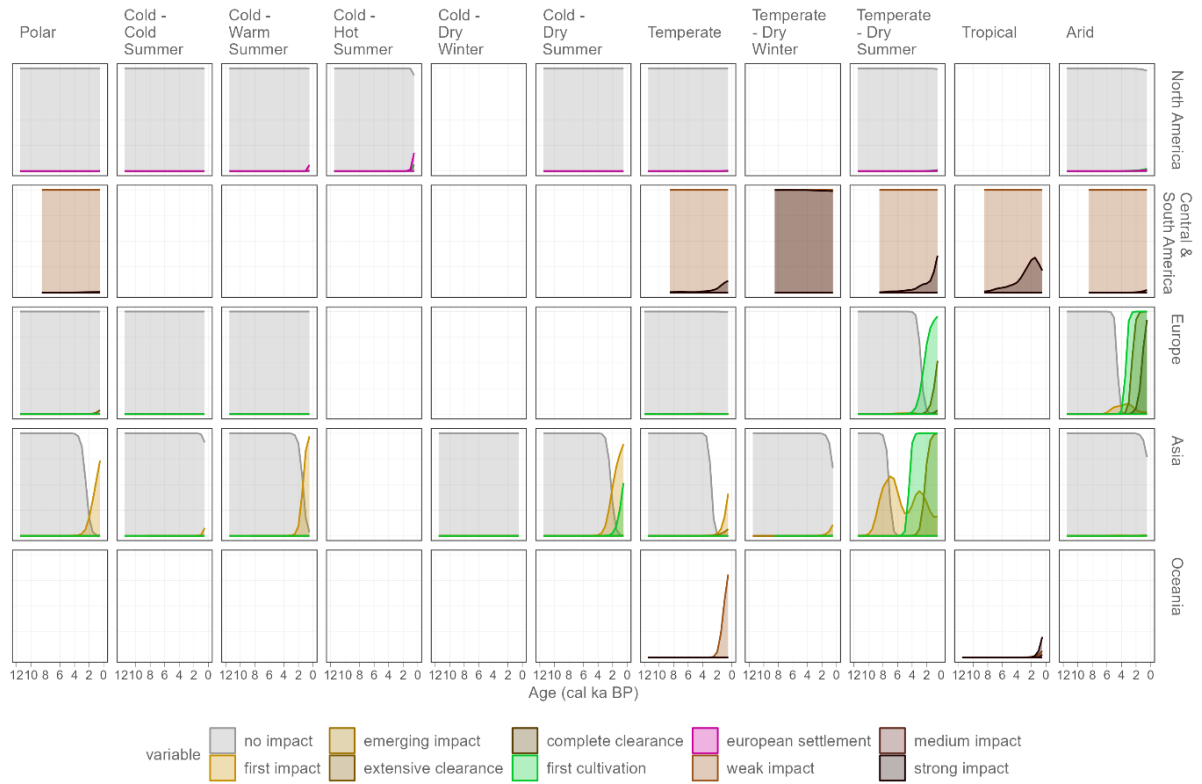
Extended Fig. 6: General patterns of minimum temperatures of the coldest month °C (*bio6*) in each climate zone and continent, derived CHELSA climate data²⁷. Thin lines represent different trends in the raw data at the sequence level. Thick lines show the general prediction from the hierarchical generalised additive models (HGAMs).



Extended Fig. 7: General patterns of precipitation of the warmest quarter (*bio18*) in each climate zone and continent, derived from CHELSA climate data²⁷. Thin lines represent different trends in the raw data at the sequence level. Thick lines show the general prediction from the hierarchical generalised additive models (HGAMs).



Extended Fig. 8: General patterns of precipitation of the coldest quarter (bio19) in each climate zone and continent, derived from CHELSA climate data²⁷. Thin lines represent different trends in the raw data at the sequence level. Thick lines show the general prediction from the hierarchical generalised additive models (HGAMs).



Extended Fig. 9: General pattern of hierarchical generalised additive models (HGAMs) showing the detection of different human events from pollen sequences summarised by climate zone for each geographical region. Different events are detected in different regions (see Extended Table 2).

Extended Table 1

Selection of settings applied in our FOSSILPOL workflow⁵¹.

Settings	Setting type	Selection
General		
	geography_criteria	-180
	long_max	180
	lat_min	-90
	lat_max	90
	alt_min	
	alt_max	
	private_data	TRUE
Neotoma		
	dataset_type	pollen
	sel_var_element	pollen
	chron_order.type1	Varve years BP
	chron_order.type2	Calibrated radiocarbon years BP
	chron_order.type3	Calendar years BP
	chron_order.type4	Radiocarbon years BP
	chron_order.type5	Calendar years AD/BC
	chron_order.type6	
Age-depth models		
	min_n_of_control_points	3
	default_thickness	TRUE
	default_error	100
	max_age_error	3000
	guess_depth	10
	default_iteration	10000
	default_burn	2000
	default_thin	8
	iteration_multiplier	5
Site filtering		
	pollensum.filter_by_pollen_sum	TRUE
	pollensum.min_n_grains	25
	pollensum.target_n_grains	150
	pollensum.percentage_samples	50
	filter_by_age_limit	TRUE
	extrapolation.filter_by_extrapolation	TRUE
	extrapolation.maximum_age_extrapolation	3000
	extrapolation.filter_by_interest_region	TRUE
	extrapolation.n_levels.filter_by_number_of_levels	TRUE

Settings	Setting type	Selection
	extrapolation.n_levels.min_n_levels	5
	extrapolation.use_age_quantiles	TRUE
	extrapolation.use_bookend_level	TRUE
Data download: 23-05-2022		

Extended Table 2
Type of human events identified in pollen sequences

Region	Human-impact event type	Detection criteria
North America	BI = No Evidence of Impact	
North America	FCu = First Cultivation	First unambiguous palynological evidence for cultivation of cereals (<i>Zea mays</i> , <i>Triticum</i> , <i>Avena</i> , etc.) and other crop plants (e.g. <i>Cannabis sativa</i>).
North America	ES = European Settlement	First unambiguous indications of European settlement as shown in eastern and central US by a marked expansion of <i>Ambrosia</i> -type pollen, often accompanied by rises of <i>Amaranthaceae</i> undiff. (= <i>Chenopodium</i> -type), weeds such as <i>Plantago</i> , <i>Salsola kali</i> , and cultivated plants such as <i>Zea mays</i> , other cereals, and <i>Cannabis sativa</i> . In the Canadian prairie, the palynological evidence is often sustained rises of <i>Cyclachaena xanthifolia</i> -type (= <i>Iva xanthifolia</i> -type), <i>Salsola kali</i> , <i>Amaranthaceae</i> undiff., and <i>Iva ciliata</i> -type. In the Pacific North West, (e.g. western Washington, western Oregon, British Columbia, northern California), palynological evidence for European settlement in the mid-1800s consists of marked declines in coniferous tree pollen (e.g. <i>Pseudotsuga (menziesii)</i> <i>Thuja (plicata)</i> , <i>Tsuga (heterophylla)</i> , <i>Abies (grandis)</i> , <i>Picea (sitchensis)</i> , and sharp rises in pollen of <i>Alnus (rubra)</i> and <i>Populus (trichocarpa)</i> .
Europe	BI = No Evidence of Impact	

Region	Human-impact event type	Detection criteria
Europe	FI = First Indication	First unambiguous indications of human impact, as shown in northern and central Europe by the beginnings of a continuous curve for <i>Plantago lanceolata</i> -type pollen. In the Alps and southern Europe, the curves for <i>P. atrata</i> -type, <i>P. major</i> -type, and/or <i>P. maritima</i> -type pollen are, in the absence of <i>P. lanceolata</i> -type, good indicators as they contain species (e.g. <i>P. major</i> , <i>P. serpentina</i>) that have very similar ecologies to <i>P. lanceolata</i> .
Europe	FCu = First Cultivation	First unambiguous indicators of cultivation of cereals and, at some sites, <i>Cannabis sativa</i> .
Europe	EC = Extensive Clearance	Onset of extensive clearance (50%) of forest within the pollen-source area (PSA) of the sites. The criteria used involve a) a synthesis of work on the AP/NAP ratio; b) use of Poaceae % values (expressed as % total tree pollen) to define extensive clearance (ca. 50% of the PSA of the site being cleared of woodland); c) use of general pollen-representation factors to determine what Poaceae % values (expressed as % total pollen excluding spores and obligate aquatics) when transformed give an inferred value of 50% forestless landscape; and d) examination of pollen stratigraphies used in Sugita-type REVEALS reconstruction of past vegetation at the landscape-scale. We use 10% Poaceae or 10% <i>Calluna</i> pollen as the criterion for EC in this work.
Europe	CC = Complete Clearance	Onset of complete clearance of forest within the PSA. The decision of what criterion to use to recognise CC follows those used to derive values for the onset of extensive clearance. The criterion is 20% Poaceae or 20% <i>Calluna</i> pollen (expressed as % total pollen excluding spores and obligate aquatics).
Asia	BI = No Evidence of Impact	

Region	Human-impact event type	Detection criteria
Asia	FI = First Indication	First unambiguous palynological indications of human impact as shown by statistically significant increases in Poaceae pollen, corresponding decreases in forest tree pollen, and the beginnings of scattered occurrences of, inter alia, <i>Plantago</i> , <i>Rumex</i> , Amaranthaceae, and Fabaceae pollen. In tropical sequences (e.g. Malayan Borneo), it is based on marked declines in pollen values of 'primary' forest trees and an increase in pollen of 'secondary' trees such as <i>Trema</i> .
Asia	FCu = First Cultivation	First unambiguous palynological evidence for cultivation of cereals and other crop plants (e.g. <i>Fagopyrum</i> , <i>Cannabis sativa</i> , <i>Juglans regia</i>).
Asia	EI = Extensive Impact	Unambiguous palynological evidence for 'extensive impact' in the Levant area as provided by massive deforestation and major changes in palynological composition.
Oceania	none = No Impact	No discernible evidence for human influence on vegetation change (note: In Oceania human interactions with vegetation has an antiquity of at least 50,000 years, however, this interaction is difficult to unequivocally detect in the pollen and charcoal records)
Oceania	weak = Weak Impact	Low impact on temperate vegetation is shown by minor increases in open vegetation (<i>Poaceae</i>), woodland disturbance (changes in dominance of fire-sensitive (e.g. <i>Callitris</i> , <i>Casuarina</i>) versus fire-adapted trees (including <i>Eucalyptus</i>), plus minor increases in fire events. In tropical sequences (e.g. Papua New Guinea), it is marked by minor declines in 'primary' forest pollen values of forest trees and minor increases in pollen of 'secondary' trees such as <i>Trema</i> and <i>Dodonaea</i> .

Region	Human-impact event type	Detection criteria
Oceania	medium = Medium Impact	Medium impact on temperate vegetation is shown by significant increases in open vegetation (<i>Poaceae</i>), woodland disturbance (e.g. prolonged declines in fire sensitive vegetation, such as long lived <i>Callitris</i> , <i>Casuarina</i> , and <i>Dicksonia</i> ferns, and an increase in fire-adapted vegetation including <i>Eucalyptus</i>), plus rapid changes in occurrence of fire events. In tropical sequences (e.g. Papua New Guinea), it is based on marked declines in 'primary' forest pollen values of forest trees and an increase in pollen of 'secondary' trees such as <i>Trema</i> and <i>Dodonaea</i> .
Oceania	strong = Strong Impact	Unambiguous palynological evidence for 'strong impact' is evidenced by a combination of massive deforestation due to clearance, by woody thickening the region due to the impact of colonisation, and a disruption of fire regimes reflected in variable fire responses. In temperate regions of Oceania (e.g. Australia) this is a result of forced cessation of Indigenous land management practices and use of fire to maintain once open landscapes at the time of colonisation around 300-240 years ago. The pollen sequences are characterised by major changes in palynological composition including introduction of weeds (e.g. <i>Pinus</i> and <i>Rumex</i>). In the tropics strong impact is evident in some sites with significant clearance of forests, increased <i>Poaceae</i> and the rise in <i>Casuarina</i> due to deliberate planting for horticultural purposes (occurs in Papua New Guinea montane sites after 2000 yr BP).

Extended Table 3

Ratio of importance of past human presence (SPD) and palaeoclimate

Age ka BP	Predictor	North America	Central & South America	Europe	Asia	Oceania
500	climate	0.66	1.00	0.87	1.00	0.92
500	human	0.34	0.00	0.13	0.00	0.08
1000	climate	0.90	1.00	0.95	0.87	0.67
1000	human	0.10	0.00	0.05	0.13	0.33
1500	climate	0.93	0.89	0.98	0.94	0.68
1500	human	0.07	0.11	0.02	0.06	0.32
2000	climate	0.95	0.98	0.96	0.79	0.26
2000	human	0.05	0.02	0.04	0.21	0.74
2500	climate	0.92	0.96	0.95	0.94	0.78
2500	human	0.08	0.04	0.05	0.06	0.22
3000	climate	0.90	0.95	0.91	0.95	0.80
3000	human	0.10	0.05	0.09	0.05	0.20
3500	climate	0.96	0.93	0.72	0.94	0.85
3500	human	0.04	0.07	0.28	0.06	0.15
4000	climate	0.90	0.88	0.93	0.94	0.90
4000	human	0.10	0.12	0.07	0.06	0.10
4500	climate	0.91	0.92	0.97	0.83	0.96
4500	human	0.09	0.08	0.03	0.17	0.04
5000	climate	0.94	0.88	0.93	0.77	0.83
5000	human	0.06	0.12	0.07	0.23	0.17
5500	climate	0.88	0.95	0.99	0.93	0.92
5500	human	0.12	0.05	0.01	0.07	0.08
6000	climate	0.88	0.94	0.86	0.75	1.00
6000	human	0.12	0.06	0.14	0.25	0.00
6500	climate	0.91	1.00	0.66	0.95	1.00
6500	human	0.09	0.00	0.34	0.05	0.00
7000	climate	0.89	0.95	0.80	0.83	1.00
7000	human	0.11	0.05	0.20	0.17	0.00
7500	climate	0.89	0.96	0.54	0.86	1.00
7500	human	0.11	0.04	0.46	0.14	0.00
8000	climate	0.95	0.70	0.84	0.94	1.00
8000	human	0.05	0.30	0.16	0.06	0.00
8500	climate	0.96	0.93	0.89	0.97	1.00
8500	human	0.04	0.07	0.11	0.03	0.00

Extended Table 4

Ratio of importance of human-impact events and palaeoclimate

Age ka BP	Predictor	North America	Central & South America	Europe	Asia	Oceania
500	climate	0.65	0.88	0.82	0.53	1.00

Age ka BP	Predictor	North America	Central & South America	Europe	Asia	Oceania
500	human	0.35	0.12	0.18	0.47	0.00
1000	climate	0.87	1.00	0.84	0.46	0.73
1000	human	0.13	0.00	0.16	0.54	0.27
1500	climate	0.94	0.99	0.82	0.40	1.00
1500	human	0.06	0.01	0.18	0.60	0.00
2000	climate	0.97	0.96	0.79	0.48	0.69
2000	human	0.03	0.04	0.21	0.52	0.31
2500	climate	0.94	0.88	0.85	0.61	0.84
2500	human	0.06	0.12	0.15	0.39	0.16
3000	climate	0.93	0.97	0.86	0.61	0.88
3000	human	0.07	0.03	0.14	0.39	0.12
3500	climate	0.92	0.89	0.63	0.90	1.00
3500	human	0.08	0.11	0.37	0.10	0.00
4000	climate	0.92	0.93	0.93	0.71	1.00
4000	human	0.08	0.07	0.07	0.29	0.00
4500	climate	0.93	0.94	0.95	0.82	NA
4500	human	0.07	0.06	0.05	0.18	NA
5000	climate	0.90	1.00	0.89	0.78	NA
5000	human	0.10	0.00	0.11	0.22	NA
5500	climate	0.90	1.00	0.97	0.88	NA
5500	human	0.10	0.00	0.03	0.12	NA
6000	climate	0.91	1.00	0.74	0.94	NA
6000	human	0.09	0.00	0.26	0.06	NA
6500	climate	0.91	0.98	0.97	0.98	NA
6500	human	0.09	0.02	0.03	0.02	NA
7000	climate	0.99	0.94	0.80	0.93	NA
7000	human	0.01	0.06	0.20	0.07	NA
7500	climate	0.86	0.98	0.75	0.95	NA
7500	human	0.14	0.02	0.25	0.05	NA
8000	climate	0.81	1.00	0.90	0.95	NA
8000	human	0.19	0.00	0.10	0.05	NA
8500	climate	0.71	1.00	0.95	0.93	NA
8500	human	0.29	0.00	0.05	0.07	NA